

Assignment – 1.5

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Batch - 04

AI Assisted Coding

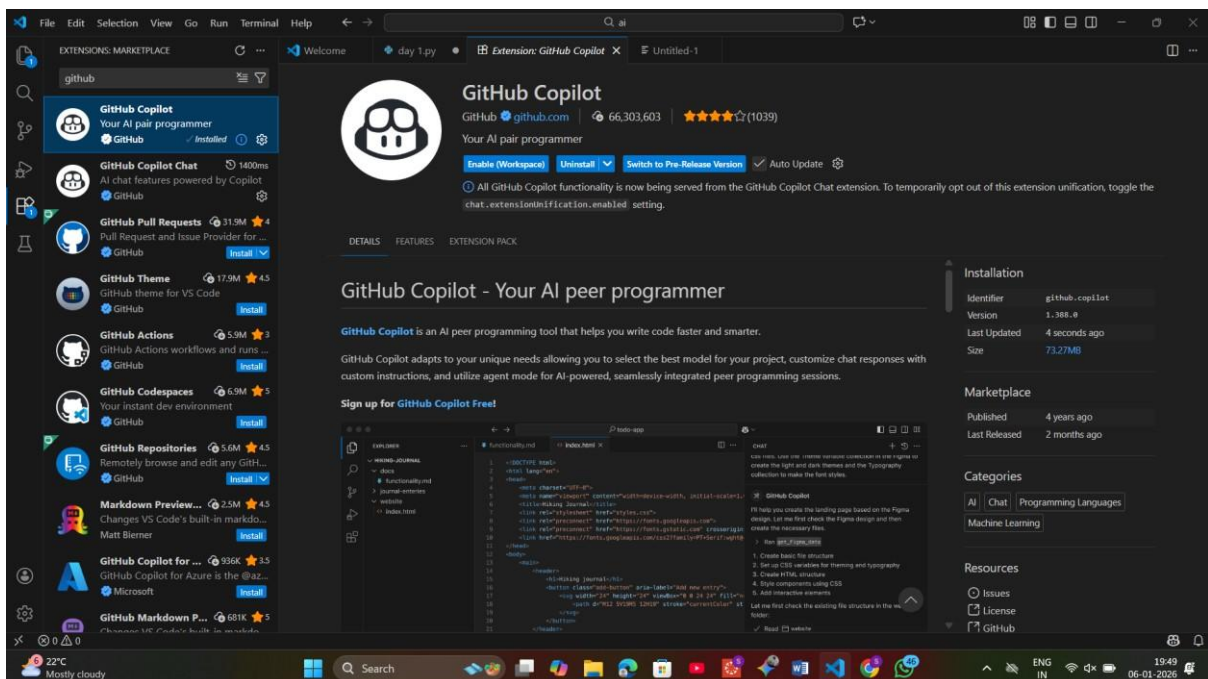
Task 0: Environment Setup:-

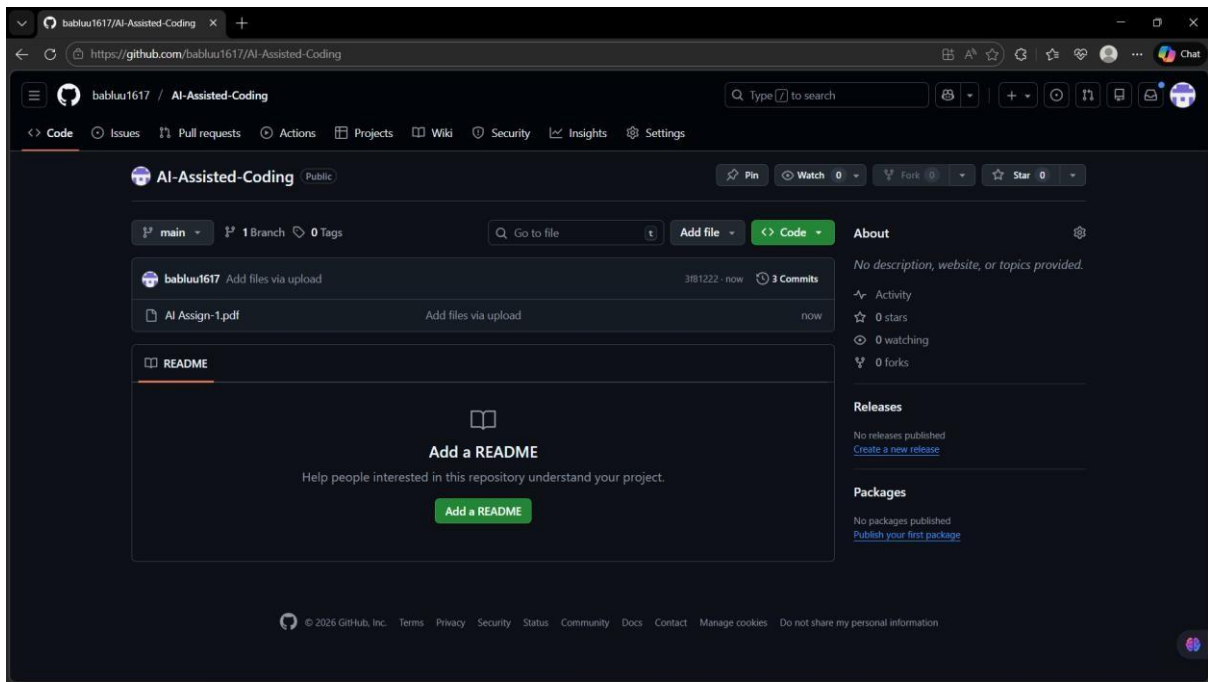
Task 0

- Install and configure GitHub Copilot in VS Code. Take screenshots of each step.

Expected Output

- Install and configure GitHub Copilot in VS Code. Take screenshots of each step.





Task 1: Non-Modular Logic (Factorial):-

: AI-Generated Logic Without Modularization (String Reversal Without Functions)

❖ Scenario

You are developing a basic text-processing utility for a messaging application.

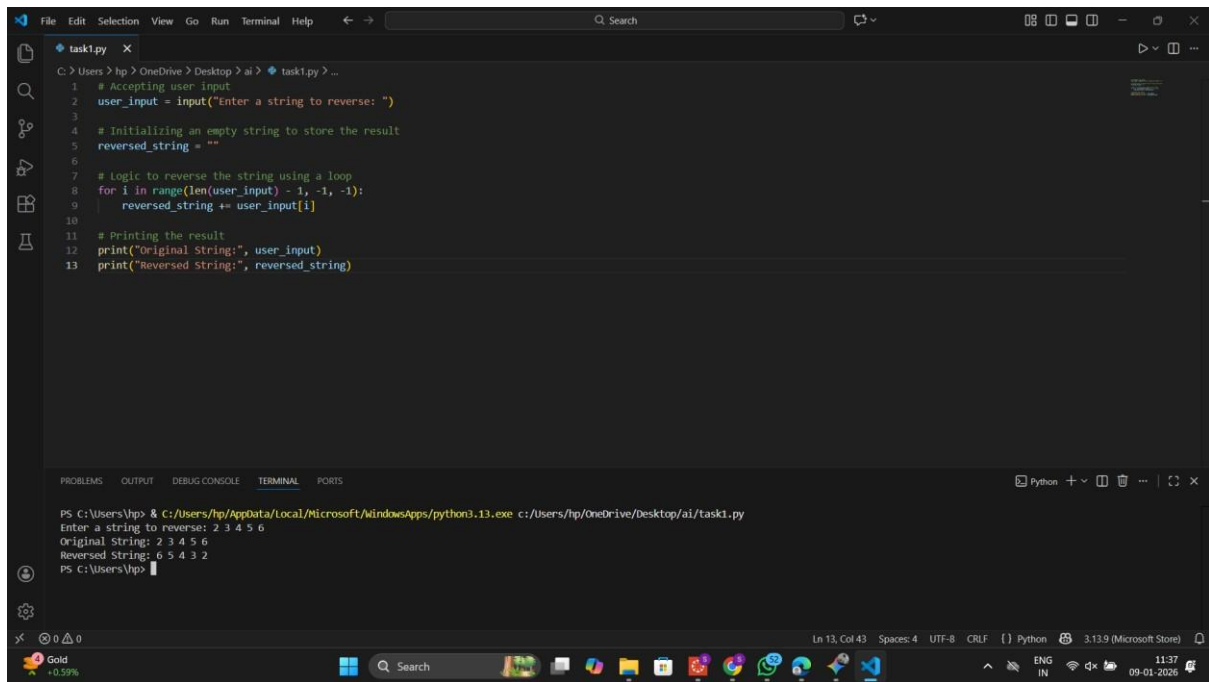
❖ Task Description

Use GitHub Copilot to generate a Python program that:

- Reverses a given string
- Accepts user input
- Implements the logic directly in the main code
- Does not use any user-defined functions

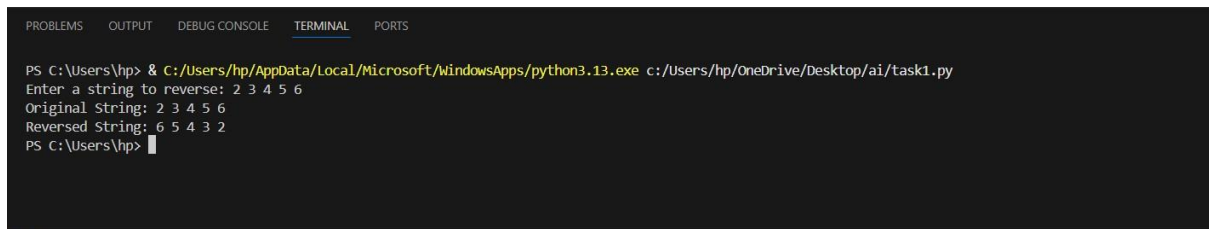
❖ Expected Output

- Correct reversed string
- Screenshots showing Copilot-generated code suggestions
- Sample inputs and outputs



```
task1.py
1 # Accepting user input
2 user_input = input("Enter a string to reverse: ")
3
4 # Initializing an empty string to store the result
5 reversed_string = ""
6
7 # Logic to reverse the string using a loop
8 for i in range(len(user_input) - 1, -1, -1):
9     reversed_string += user_input[i]
10
11 # Printing the result
12 print("Original String:", user_input)
13 print("Reversed String:", reversed_string)
```

```
PS C:\Users\hp> & C:/Users/hp/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/hp/OneDrive/Desktop/ai/task1.py
Enter a string to reverse: 2 3 4 5 6
Original String: 2 3 4 5 6
Reversed String: 6 5 4 3 2
PS C:\Users\hp>
```



```
PS C:\Users\hp> & C:/Users/hp/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/hp/OneDrive/Desktop/ai/task1.py
Enter a string to reverse: 2 3 4 5 6
Original String: 2 3 4 5 6
Reversed String: 6 5 4 3 2
PS C:\Users\hp>
```

Task 2: AI Code Optimization:-

Efficiency & Logic Optimization (Readability Improvement)

❖ Scenario

The code will be reviewed by other developers.

❖ Task Description

Examine the Copilot-generated code from Task 1 and improve it by:

- Removing unnecessary variables
- Simplifying loop or indexing logic
- Improving readability
- Use Copilot prompts like:

▪ “Simplify this string reversal code” ▪

“Improve readability and efficiency”

Hint:

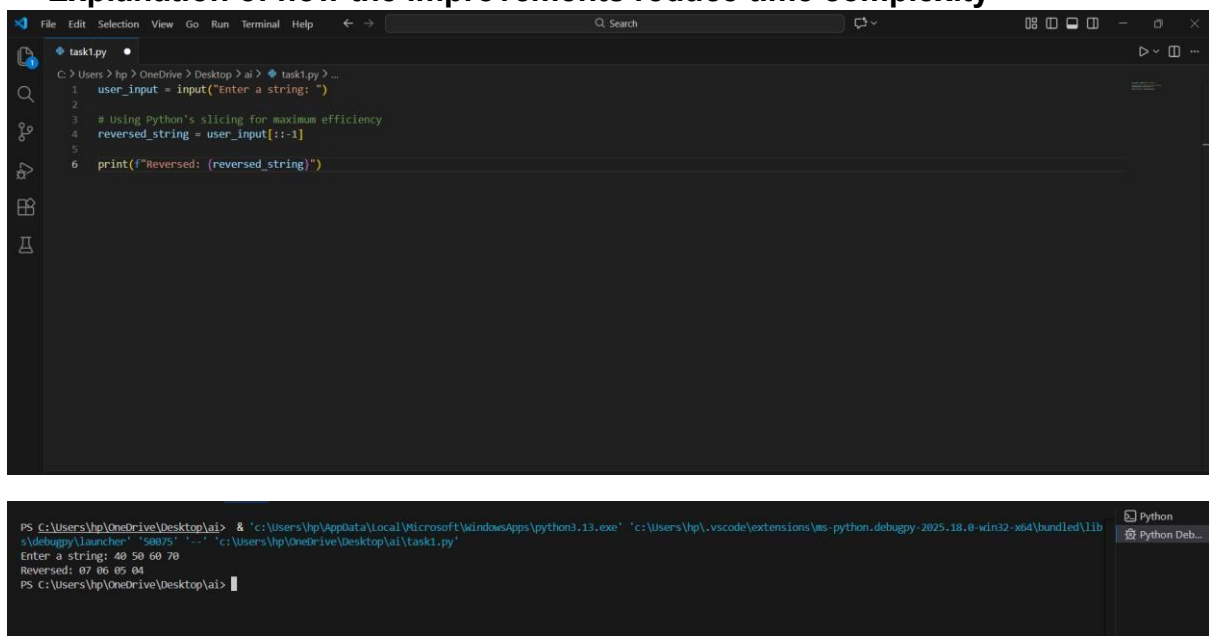
Prompt Copilot with phrases like

“optimize this code”, “simplify logic”, or “make it more readable”

❖ Expected Output

➤ Original and optimized code versions

➤ Explanation of how the improvements reduce time complexity



The image shows a screenshot of a Visual Studio Code editor window. The editor is open to a file named `task1.py`. The code in the file is as follows:

```
C:\Users\hp> OneDrive\Desktop> ai> task1.py > -
1 user_input = input("Enter a string: ")
2
3 # Using Python's slicing for maximum efficiency
4 reversed_string = user_input[::-1]
5
6 print(f"Reversed: {reversed_string}")
```

Below the editor, a terminal window shows the execution of the script. The prompt is `PS C:\Users\hp\OneDrive\Desktop\ai>`. The command `& 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\lib\debugpy\launcher' '58075' '-c' 'c:\Users\hp\OneDrive\Desktop\ai\task1.py'` is entered. The output shows the prompt `Enter a string: 40 50 60 70`, followed by `Reversed: 07 06 05 04`. The terminal prompt returns to `PS C:\Users\hp\OneDrive\Desktop\ai>`.

Task 3: Modular Design Using AI Assistance (String Reversal Using Functions) ❖

Scenario

The string reversal logic is needed in multiple parts of an application.

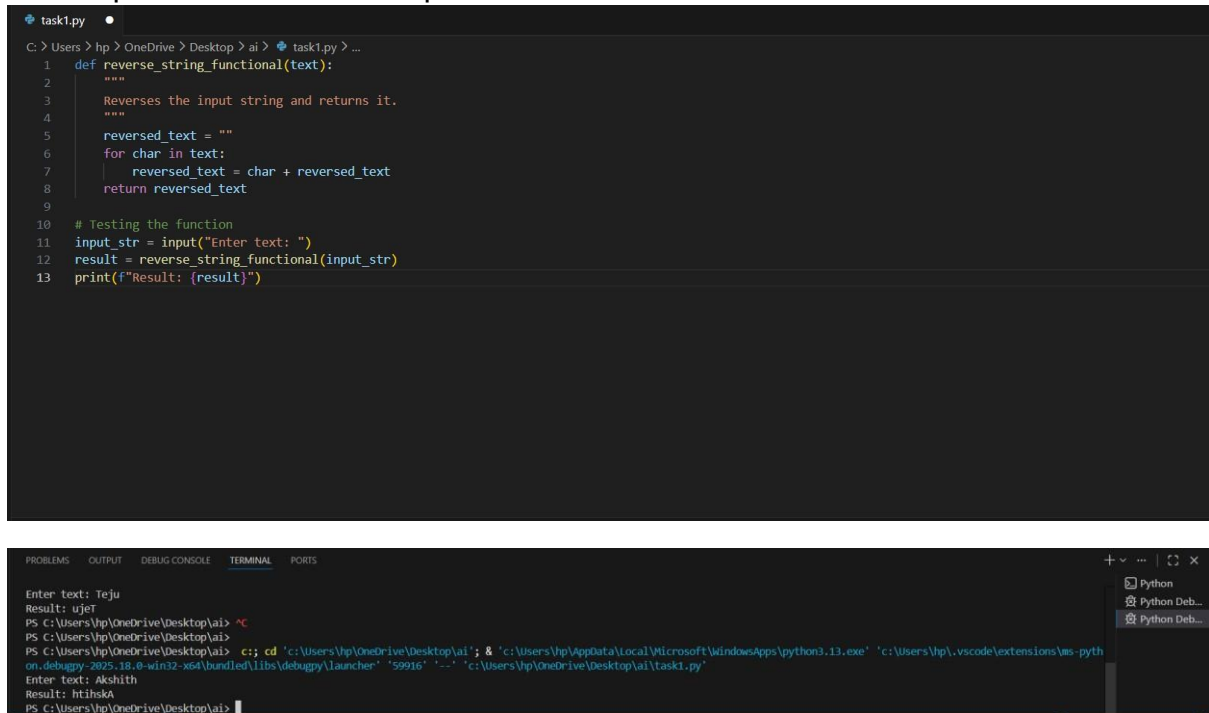
❖ Task Description

Use GitHub Copilot to generate a function-based Python program that:

- Uses a user-defined function to reverse a string
- Returns the reversed string
- Includes meaningful comments (AI-assisted)

❖ Expected Output

- Correct function-based implementation
- Screenshots documenting Copilot's function generation
- Sample test cases and outputs



```
task1.py
C: > Users > hp > OneDrive > Desktop > ai > task1.py > ...
1 def reverse_string_functional(text):
2     """
3     Reverses the input string and returns it.
4     """
5     reversed_text = ""
6     for char in text:
7         reversed_text = char + reversed_text
8     return reversed_text
9
10 # Testing the function
11 input_str = input("Enter text: ")
12 result = reverse_string_functional(input_str)
13 print(f"Result: {result}")

Enter text: Teju
Result: ujeT
PS C:\Users\hp\OneDrive\Desktop\ai> ^C
PS C:\Users\hp\OneDrive\Desktop\ai>
PS C:\Users\hp\OneDrive\Desktop\ai> c:; cd 'c:\Users\hp\OneDrive\Desktop\ai'; & 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\.vscode\extensions\ms-pyth
on.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '59916' '-' 'c:\Users\hp\OneDrive\Desktop\ai\task1.py'
Enter text: Akshith
Result: htshkA
PS C:\Users\hp\OneDrive\Desktop\ai>
```

Task 4: Comparative Analysis – Procedural vs Modular Approach (With vs Without Functions)

❖ Scenario

You are asked to justify design choices during a code review.

❖ Task Description

Compare the Copilot-generated programs:

- Without functions (Task 1)
- With functions (Task 3) Analyze them based on:
 - Code clarity
 - Reusability
 - Debugging ease

➤ **Suitability for large-scale applications**

❖ **Expected Output**

Comparison table or short analytical report

Feature	Procedural (Without Functions)	Modular (With Functions)
Code Clarity	Easy for tiny scripts, but harder for large ones.	Very high; logic is isolated and named.
Reusability	Must copy-paste code if reused, leading to redundancy.	Can be called anywhere in the app.
Debugging	Harder to isolate where an error occurs.	Easy to unit test the specific function.
Scalability	Not suitable for large-scale applications.	Essential for professional development.

Task 5: AI-Generated Iterative vs Recursive Fibonacci Approaches (Different Algorithmic Approaches to String Reversal)

❖ **Scenario**

Your mentor wants to evaluate how AI handles alternative logic paths.

❖ **Task Description**

Prompt GitHub Copilot to generate:

➤ **A loop-based string reversal approach**

➤ **A built-in / slicing-based string reversal approach**

❖ **Expected Output**

➤ **Two correct implementations**

➤ **Comparison discussing:**

▪ **Execution flow**

▪ **Time complexity**

- Performance for large inputs
- When each approach is appropriate.

The image shows a screenshot of the Visual Studio Code editor. The top panel displays a Python file named `task1.py` with the following code:

```
1 def reverse_iterative(input_string):
2     reversed_str = ""
3     for char in input_string:
4         reversed_str = char + reversed_str
5     return reversed_str
6
7 def reverse_slicing(input_string):
8     return input_string[::-1]
9
10 test_input = input("Enter a string: ")
11
12 print(reverse_iterative(test_input))
13 print(reverse_slicing(test_input))
```

The bottom panel shows the terminal output. The command prompt is at `C:\Users\hp\OneDrive\Desktop\ai>`. The user has run `python task1.py`, which prompts for input. The user enters `1 2 3 4 5`. The output shows the string reversed using both methods:

```
s\debuggy\launcher' '50436' '-' 'c:\Users\hp\OneDrive\Desktop\ai\task1.py'
PS C:\Users\hp\OneDrive\Desktop\ai> ^C
PS C:\Users\hp\OneDrive\Desktop\ai>
PS C:\Users\hp\OneDrive\Desktop\ai> c;; cd 'c:\Users\hp\OneDrive\Desktop\ai'; & 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\OneDrive\Desktop\ai\task1.py'
Enter a string: 1 2 3 4 5
5 4 3 2 1
5 4 3 2 1
PS C:\Users\hp\OneDrive\Desktop\ai>
```

The status bar at the bottom indicates the file is at line 13, column 35, with 4 spaces, UTF-8 encoding, CRLF line endings, and the Python 3.13.0 interpreter is selected.